Precept 2

Goals

- Review Bayes formula and its use to compute conditional probabilities
- See how to solve story problems and apply this method

Reminder (**Conditional Probabilities, Independence and Bayes formula**)

- The **conditional probability** $\mathbb{P}(A|B)$ of an event A given that B occurs is defined as :

$$\mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

provided that $\mathbb{P}(B) > 0$. Hence, $\mathbb{P}\{A \cap B\} = \mathbb{P}\{A|B\} \cdot \mathbb{P}\{B\}$.

- The events A and B are **independent** if

$$\mathbb{P}(A|B) = \mathbb{P}(A) \text{ or equivalently } \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

For **independent** events $A_1, A_2, \ldots, A_k$, we have $\mathbb{P}\{A_1 \cap A_2 \cap \ldots \cap A_k\} = \mathbb{P}\{A_1\} \times \ldots \times \mathbb{P}\{A_k\}$.

- **Law of total probability** : $\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^C) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B^C)$

Generalized Law of Total Probability : Let $\{A_i\}_{i=1,2,\ldots}$ be a **finite or countably infinite partition** of the sample space Ω, then for an event E defined on the probability space, we have :

$$\mathbb{P}(E) = \sum_i \mathbb{P}(E \cap A_i) = \sum_i \mathbb{P}(E|A_i)\mathbb{P}(A_i).$$

- **Bayes formula** :

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A|B)\mathbb{P}(B)}{\mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^C)\mathbb{P}(B^C)}.$$

Exercises

Exercise 1 (Monty Hall Problem)

This problem is about a fictional game show in which you, the player, pick a door and get whatever is behind it. One door has a shiny new sports car, the other two have goats. Suppose that you're on this show and that after you make your pick the host (Monty Hall) opens one of the other doors to reveal a goat. Before your chosen door is opened you have the option of switching to the other, yet unopened door. Do you take it?

Exercise 2 (Communication through a Noisy Channel)

A source transmits a message (a string of symbols) through a noisy communication channel. Each symbol is 0 or 1 with probability p and $1 - p$, respectively. And each symbol is received incorrectly with probability ϵ_0 and ϵ_1, respectively (see Figure 1). Errors in different symbol transmissions are independent.

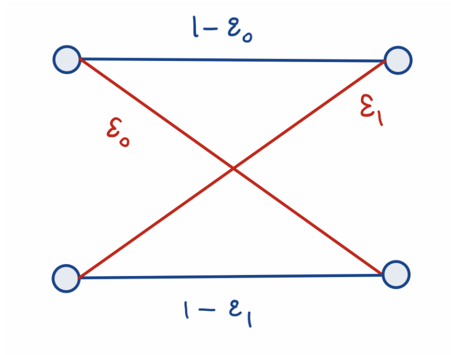

Figure 1: Probabilities in a binary communication channel

(a) What is the probability that the k-th symbol is received correctly?

(b) What is the probability that the string of symbols 1011 is received correctly?

(c) To improve reliability, each symbol is transmitted three times, and the received string is decoded by majority rule. In other words, a 0 (or 1) is transmitted as 000 (or 111) and it is decoded at the receiver if and only if the received three-symbol string contains at least two 0s (or two 1s). What is the probability that a 0 is correctly decoded?

(d) Suppose the three-symbol scheme is used. What is the probability that a symbol was 0 given that the received string is 101?

Exercise 3 (Taxi Cab Paradox) - *time-permitting*

Overnight a Taxi in New York City is involved in a hit-and-run accident. There are two taxi companies in the city, one with yellow cars and one with green cars. The yellow cab company has 85% of taxis in the city and the green cab company has 15%. There was one witness to the hit-and-run, who thought that the taxi was green. In order to determine the reliability of the witness the court ordered a test and found that the witness correctly identified the color of the taxi 80% of the time, and got it wrong 20% of the time. What is the probability that the taxi involved in the hit-and-run accident was green?